

His Uncle, Gambling, and Murder: Phonological Adaptation of Southern Min Loanwords in Cebuano

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1 Introduction

Southern Min (閩南語), also known as Hokkien, has lasted an impactful influence on the lexicons of several Southeast Asian languages through centuries of Chinese merchant migration, such as the huge amount of migration during the period of late Qing dynasty and Republican periods, known as *Xia NanYang* (下南洋) in Chinese context. In the Philippines, this influence is most directly documented in Tagalog, where Chan-Yap (1977) identified 163 Hokkien loanwords covering the semantic ranges of kinship, Southern Min food, commerce, and domestic life. What is striking is that a bunch of words that represent the uniqueness of Southern Min culture were adopted to Tagalog and Cebuano. For example, [ín-chek] (Southern Min for “his uncle”) became [in'tsik] in many Philippine languages, including Cebuano; this likely reflects intermarriage between Southern Min merchants and local women, which carried kinship terms into the local languages alongside commercial ones. Another example is that the jargon used for gambling in Southern Min were used in Philippine languages as well, such as [pak⁴.kiau⁵³] (Southern Min) and [pak⁷kjaw] (Cebuano). Furthermore, the expression of “to murder” in Southern Min is ‘pasilang’, literally meaning to fight someone till the person dies, is also frequently used in the Philippines, such as in official news. However, the phonological mapping rules of Hokkien-derived loanwords in Philippine languages has received comparatively little systematic analysis. This paper, grateful to the help of a Cebuano speaker, via an experimental phonology way, focus in the investigation of Phonological Adaptation of Southern Min Loanwords in Cebuano, the most widely spoken Austronesian language in the Philippines, with over 20 million speakers concentrated in the Visayas and Mindanao regions (Shryock 1993: p. 103).

This paper documents and analyzes the phonological adaptation of Hokkien loanwords into Cebuano, drawing on a corpus of 65 lexical items based on the wordlist of Hokkien loanwords in Tagalog (Chan-Yap 1977) and elicited from CEB001, a native Cebuano speaker. The reason for using the wordlist documented by Chan-Yap is because there are many common Southern Min loanwords in Tagalog and Cebuano, which lays a solid foundation for this case study. As a result, fourteen major phonological rules are identified and formalized, covering consonant adaptation (nasalization loss, nasal place assimilation, aspiration, /ts/ replacement, glottal stop alternations, coda stop preservation), vowel adaptation (ɔ→o substitution, glide formation, diphthong restructuring, vowel raising, and lowering), and morpheme-level processes (truncation, epenthesis, and reduplication).

This research further argues that certain phonological phenomena, such as the observed variation in aspiration and affricate replacement, is best ex-

plained by positing two historical routes of borrowing: (a) direct Southern Min-to-Cebuano borrowing, and (b) indirect borrowing via Tagalog. Meanwhile, this hypothesis accounts for several otherwise puzzling exceptions in the data. The structure of the paper is as follows: Section 2 describes the data collection methodology; Sections 3–5 constitute the core results and discussion, organized by phonological domain; Section 6 provides a synthesis and rule summary; Section 7 concludes with directions for future research; and the full lexical corpus is provided in Appendix A, following the Works Cited and AI Usage Report.

2 Methodology

Data were collected through structured elicitation sessions with CEB001, a native speaker of Cebuano. The elicitation proceeded in three stages.

First, a wordlist of 65 candidate Hokkien-origin lexical items was compiled by cross-referencing the Tagalog loanword inventory in Chan-Yap (1977) with standard Cebuano dictionaries, notably Wolff (1972) and Bunye & Yap (1971). There are certain words that do not exist in dictionaries, relying on the native check by CEB001. The Hokkien source forms were verified against the Ministry of Education Taiwanese Hokkien online dictionary (sutian.moe.edu.tw) and the Revised Formosan Pronunciation Dictionary (chhoe.taigi.info), both of which provide standardized IPA transcriptions with tone notation for citation forms.

Second, NAN001 (the author of this paper), a native Southern Min speaker, served as a secondary consultant to confirm that the proposed source forms fell within the productive lexicon of Southern Min. The validity of using Taiwanese Southern Min dictionaries is because (a) which is the only available and accessible systematically organized dictionary; (b) Taiwanese Southern Min represents the mixture of Quanzhou and Zhangzhou accents, resembling the accents of migration to the Philippines.

Third, CEB001 was asked to produce each lexical item in citation form, and recordings were made. Phonetic transcriptions were produced by the researcher using IPA notation based on close listening, with particular attention to the realization of coda stops, aspiration contrasts, stress placement, and vowel quality. Cebuano IPA forms were cross-checked against Binisaya.com and Wiktionary where available. The transcriptions represent CEB001's production and may not generalize to all Cebuano dialects or speakers.

The resulting corpus was analyzed by formulating phonological rules in a descriptive format, grouping them into three domains: consonant-relevant rules, vowel-relevant rules, and morpheme-level rules. Where a rule has counterexamples in the data, these are flagged explicitly. Tagalog comparanda are drawn from Chan-Yap (1977) and serve a comparative rather than primary analytical role in this research. The full corpus is presented in Appendix A; in-text examples throughout Sections 3–5 are drawn from and consistent with that corpus.

3 Results and Discussion: Consonant-Relevant Rules

3.1 Nasalization Loss

In Southern Min (Hokkien), nasalized vowels ($\tilde{V}$) contrast phonemically with their oral counterparts and can form minimal pairs. For example, [suã²¹] ‘mountain’ is distinct from [sua²¹] ‘sand’. Cebuano, by contrast, lacks phonemic vowel nasality: nasalized vowels do not exist in the Cebuano phoneme inventory and therefore do not form minimal pairs. Consequently, Southern Min loanwords consistently undergo nasalization loss in adaptation.

Across all examples [1.(1)–(7)], nasality is categorically dropped. Aryanti (2021: p. 255) documents the same process in Hokkien-to-Indonesian loanword adaptation, noting that vowel nasality is not contrastive in Indonesian and is therefore lost. The same filter applies in Cebuano.

Rule 1.1: $\tilde{V} \rightarrow V$

	Southern Min (Hokkien)	Cebuano
(1)	[tĩ ⁵⁵ .kue ⁵³]	[ti'koj]
(2)	[tsʰĩ ⁵⁵ .tau ³³]	[si'taw]
(3)	[tau ³³ .sĩ ³³]	[taw'si]
(4)	[ho ⁵³ .piã ⁵³]	[hokʰ'pja]
(5)	[dzun ³³ .piã ⁵³]	[dzun'pja]
(6)	[mi ³³ .suã ²¹]	[mis'wa]
(7)	[tau ³³ .kuã ⁵⁵]	[tokʰ'kwa]

3.2 Nasal Place Assimilation

Southern Min disyllabic words are composed of two morphologically independent Chinese characters, and their internal syllable boundaries are perceptually salient to speakers of Sinitic languages. In Cebuano, however, these words are borrowed as monomorphemic units, with the original morpheme boundary reanalyzed as a simple intervocalic consonant cluster. This reanalysis accelerates assimilation at the first-to-second syllable boundary, since speakers are no longer required to maintain the segmental identity of each constituent morpheme.

Four nasal place assimilation rules are attested. The general environment is: in a disyllabic word of the shape (C)VC.CV(C), when the nasal coda of the first syllable is followed by a stop onset of a different place of articulation, the nasal assimilates to the place of the following stop.

This process is typologically well-attested in loanword phonology and has been described as a form of regressive place assimilation conditioned by the following obstruent (cf. Aryanti 2021: p. 256; Kang & Katsuda 2024). Crucially, as example 2.(4) demonstrates, assimilation does not apply when the following segment is not a stop, confirming that the environment is strictly restricted to [+stop] onsets.

Rule 2.1: $n \rightarrow \eta / _ [+stop, +velar]$

	Southern Min (Hokkien)	Cebuano
(1)	[sin ⁵⁵ .k ^h eɿ ³²]	[sin'ki]

Rule 2.2: $\eta \rightarrow n$ / _ [+stop, +alveolar]

	Southern Min (Hokkien)	Cebuano
(2)	[laŋ ²⁴ .sŋ ²⁴]	[lan'soŋ]
(3)	[taŋ ²⁴ .soɿ ³²]	[tan'so]
(4)*	[sin ⁵⁵ .li ⁵³ .laŋ ²⁴]	[saŋ'laŋ]

Rule 2.3: $\eta \rightarrow m$ / _ [+stop, +bilabial]

	Southern Min (Hokkien)	Cebuano
(5)	[hoŋ ³³ .baɿ ³²]	[hum'ba]

Rule 2.4: $n \rightarrow m$ / _ [+stop, +bilabial]

	Southern Min (Hokkien)	Cebuano
(6)	[bin ³³ .po ²¹]	[bim'po]
(7)	[lun ³³ .piã ⁵³]	[lum'pja]

Rules 2.3 and 2.4 can be unified as: [+nasal] $\rightarrow m$ / _ [+stop, +bilabial]. A noteworthy counterexample is found in example 2.(8):

	Southern Min (Hokkien)	Cebuano
(8)*	[liam ²⁴ .tɔ ³³ .baɿ ³²]	[ljem'po]

Here, the expected output by Rule 2.4 would be [ljen'to] (assimilation of /m/ $\rightarrow$ /n/ before alveolar /t/). Instead, the reverse occurs: stop place assimilation applies, yielding bilabial /p/ from /t/ under the influence of the preceding nasal /m/. This reversed directionality that stops assimilating to nasal rather than nasal to stop occurs only once in the corpus, suggesting that nasal-to-stop assimilation is the dominant and more productive process, instead of a completely universal rule.

3.3 Aspiration and Deaspiration

Aspirated stops [p^h, t^h, k^h, ts^h] contrast phonemically with their unaspirated counterparts in Southern Min. In Tagalog, aspiration is not phonemic, and all Hokkien aspirated stops are deaspirated categorically upon borrowing (Chan-Yap 1977: pp. 18–19). Cebuano presents a more complex picture: aspiration is partially phonemic, with minimal pairs documented for /t/ vs. /t^h/ and /p/ vs. /p^h/ (CEB001, personal communication), while /k^h/ is absent from the Cebuano consonant inventory.

This asymmetry produces three distinct outcomes: (a) /k^h/ is categorically deaspirated (Rule 3.1) because Cebuano lacks /k^h/ entirely; (b) /t^h/ is preserved where it contrasts (Rule 3.2); and (c) the behavior of /ts^h/ and /p^h/ is variable, conditioned by the historical borrowing path. The hypothesis that two borrowing routes coexist, one directly from Southern Min, one via Tagalog, was based on the lexical overlapping of Southern Min loanwords in Tagalog and Cebuano, and the present data suggest that the dual-route model is necessary for Cebuano. In other words, without this dual-route model, the phonological explanation is insufficient to cover the variations of loanwords' pronunciation affected by historical and lexical interactions.

Rule 3.1: k^h → k

	Southern Min (Hokkien)	Cebuano
(1)	[bak ²⁴ .k ^h iaɿ ²⁴]	[bak ¹ kia]
(2)	[sin ⁵⁵ .k ^h eɿ ³²]	[siŋ ¹ ki]
(3)	[ka ³³ .i ⁵⁵ .t ^h ai ²⁴]	[ka ¹ tay]
(4)	[tsu ⁵³ .k ^h eɿ ³²]	[su ¹ ki]

Rule 3.2: t^h → t^h (aspiration preserved)

	Southern Min (Hokkien)	Cebuano
(5)	[t ^h aŋ ²⁴ .a ⁵³]	[t ^h aŋ ¹ a]
(6)	[pɔ ⁵³ .t ^h au ²⁴]	[put ¹ t ^h aw]

Rule 3.3*: ts^h → ts^h (aspiration preserved)

	Southern Min (Hokkien)	Cebuano
(7)	[peɿ ²⁴ .ts ^h ai ²¹]	[p ^h e ¹ ts ^h ay]
(8)*	[ts ^h i ⁵⁵ .tau ³³]	[si ¹ taw]

Example 3.(8) is an exception to Rule 3.3: [si¹taw] shows both deaspiration and affricate simplification (/ts^h/ → /s/). This is consistent with the hypothesis that [si¹taw] entered Cebuano via Tagalog (where /ts^hi⁵⁵/ → /si/), rather than through direct Southern Min-to-Cebuano borrowing. Rule 3.3 is marked with an asterisk because it is not universal: example 3.(8) shows that the aspirated affricate is not always preserved, conditioned by the borrowing route just discussed.

Rule 3.4: p ↔ p^h (aspiration unstable)

	Southern Min (Hokkien)	Cebuano
(7)	[peɿ ²⁴ .ts ^h ai ²¹]	[p ^h e ¹ ts ^h ay]
(9)	[p ^h aɿ ³² .si ⁵³ .laŋ ²⁴]	[pas ¹ laŋ]

The additional mystery of /p/ → /p^h/ in example 3.(7) ([peɿ²⁴.ts^hai²¹] → [p^he¹ts^hay]) and /p^h/ → /p/ in example 3.(9) ([p^haɿ³².si⁵³.laŋ²⁴] → [pas¹laŋ])

remains unresolved. The deaspiration in 3.(7) is plausibly a Tagalog-route item, but the aspiration addition in 3.(9) lacks a clear phonological conditioning environment and requires further investigation.

3.4 /ts/ Replacement

The affricate /ts/ is absent from the native Cebuano consonant inventory. Southern Min /ts/ is therefore subject to substitution upon borrowing, with the specific output segment conditioned by the borrowing route and phonological environment. Four replacement patterns are attested:

Rule 4.1: ts, ts^h → s [Tagalog-route]

	Southern Min (Hokkien)	Cebuano
(1)	[tsu ⁵³ .k ^h eɿ ³²]	[su'ki]
(2)	[gin ²⁴ .tsam ⁵³]	[gin'sam]
(3)	[ts ^h i ⁵⁵ .tau ³³]	[si'taw]

Rule 4.2: ts → ʃ [Tagalog-route; environment: _i+V]

	Southern Min (Hokkien)	Cebuano
(4)	[tsui ⁵³ .kui ⁵³]	[ʃo'koy]

Rule 4.3: ts → t [Tagalog-route]

	Southern Min (Hokkien)	Cebuano
(5)	[a ⁵⁵ .tsi ⁵³]	[a.te]

Rule 4.4: ts → ts^h [direct Southern Min route]

	Southern Min (Hokkien)	Cebuano
(6)	[baɿ ³² .tsui ⁵³]	[bat ^h 'ts ^h oj]
(7)	[dzi ³³ .tsi ⁵³]	[di'ts ^h e]

The contrast between Rule 4.3 and Rule 4.4 provides compelling evidence for the dual-route hypothesis. Both [a.te] ‘elder sister’ and [di'ts^he] ‘second elder sister’ derive from Southern Min /tsi/ and share the meaning ‘sister’, yet they surface with different consonants in Cebuano. If all Southern Min loanwords entered Cebuano via a single path, words with identical source segments and similar meanings would be expected to surface identically. The divergence strongly suggests that [a.te] passed through Tagalog (where /ts/ → /t/), while [di'ts^he] was borrowed directly from Southern Min (where /ts/ was adapted as aspirated /ts^h/, a segment that exists in Cebuano).

It is noteworthy that Rule 3.3 and Rule 4.4 together indicate an asymmetry in the Cebuano consonant inventory itself, not just in the borrowing

path: the plain affricate /ts/ does not exist in Cebuano, but its aspirated counterpart /ts^h/ does. This is why direct Southern Min borrowings preserve /ts/ as /ts^h/ (Rule 4.4) rather than as /ts/, while Tagalog-route borrowings, which lack aspiration altogether, resolve /ts/ to /s/ or /t/ instead (Rules 4.1, 4.3).

3.5 Glottal Stop Deletion and Insertion

Southern Min checked-tone syllables bear a final glottal stop /ʔ/ which is deleted in Cebuano. There is only one environment for this deletion: the glottal stop is removed whenever it occupies the coda of a syllable, regardless of whether that syllable is the first or the second syllable of the word. The deletion is categorical across the corpus. A superficially paradoxical insertion rule also exists: glottal stop is inserted between two adjacent vowels (/V_V/) because hiatus (a sequence of two tautosyllabic vowels) is disallowed in Cebuano phonotactics.

Rule 5.1: ʔ → Ø / _]σ (coda deletion)

	Southern Min (Hokkien)	Cebuano
(1)	[p ^h aʔ ³² .si ⁵³ .laŋ ²⁴]	[pas'laŋ]
(2)	[peʔ ⁴ .ts ^h ai ²¹]	[p ^h e'ts ^h ay]
(3)	[baʔ ³² .mi ³³]	['ma.mi]
(4)	[hoŋ ³³ .baʔ ³²]	[hum'ba]
(5)	[hiaʔ ⁴]	[hi'ja]
(6)	[bak ⁴ .k ^h iaʔ ⁴]	[bak ¹ 'kia]

Rule 5.2*: Ø → ʔ / V_V (hiatus repair)

	Southern Min (Hokkien)	Cebuano
(7)	[bi ⁵³ .lau ³³]	[bi'laʔ.o]

Rule 5.2 is marked with an asterisk because it is not universally applicable in the corpus: only a single example of vowel-hiatus repair was found, so the rule is best treated as conditional rather than fully general pending further data.

The contrast between glottal deletion and glottal insertion is not paradoxical but complementary: glottal stop is removed when it would produce an unacceptable coda environment within the Southern Min checked syllable, and inserted when vowel hiatus would otherwise arise.

3.6 Coda Stop Preservation and Insertion

One of the most typologically salient features of CEB001's Cebuano loan-word phonology is the preservation of Hokkien checked-tone coda stops (/k̚/, /t̚/) as unreleased stops in the Cebuano output. This stands in sharp contrast to Tagalog, where the same coda stops are deleted (Chan-Yap 1977: pp. 19–20).

Rule 6.1: $C' \rightarrow \bar{C}$ (coda stop preservation)

	Southern Min (Hokkien)	Cebuano
(1)	[pian ³³ .sit ⁴]	[pan'sit̚]
(2)	[ki ²¹ .sut̚ ³²]	['ku.sot̚]
(3)	[hok ⁷⁴ .bu ³³]	['hok̚.bo]
(4)	[pak ⁷⁴ .kiau ⁵³]	[pak̚'kjaw]
(5)	[bak ⁷⁴ .k ^h iaŋ ⁴]	[bak̚'kia]

Unreleased coda stop insertion is also attested in three items (6.(6)–(8)), though the conditioning environment is not fully transparent from the present data. Two tentative sub-environments emerge: (i) insertion of /k̚/ before a following /k/ when the preceding vowel is [+back] (6.(6)); and (ii) insertion of /t̚/ before an aspirated consonant (6.(7)–(8)).

Rule 6.2: $\emptyset \rightarrow \bar{C} / _ .C$ (coda stop insertion, tentative)

	Southern Min (Hokkien)	Cebuano
(6)	[tau ³³ .kuã ⁵⁵]	[tok̚'kwa]
(7)	[baŋ ³² .tsui ⁵³]	[bat̚'ts ^h oj]
(8)	[pɔ ⁵³ .t ^h au ²⁴]	['put̚.t ^h aw]

While Cebuano possesses both /k̚/ and /t̚/ as phonotactically licit coda stops, /p̚/ is not attested in the present data. Whether this reflects a genuine gap in the loanword stratum or simply a gap in the present corpus requires further investigation with additional elicitation items.

4 Results and Discussion: Vowel-Relevant Rules

4.1 ɔ → o Substitution

Southern Min /ɔ/ is a low-mid back rounded vowel. Cebuano's vowel inventory comprises five vowels: /i, e, a, o, u/. The Southern Min /ɔ/ is mapped to Cebuano /o/, the closest native approximation, across the corpus.

The /ɔ/→/o/ mapping is also attested in Indonesian Hokkien loanwords, where /ɔ/ is similarly absent from the target vowel inventory (Aryanti 2021: p. 258).

Rule 7: ɔ → o

	Southern Min (Hokkien)	Cebuano
(1)	[ɔ ⁵⁵ .pu ²⁴]	['u.po]
(2)	[ɔ ³³ .kue ⁵³]	[u'koj]
(3)	[gu ²⁴ .tɔ ³³]	[go'to]

4.2 Glide Formation and Diphthong Restructuring

High vowels /i, u/ are realized as the corresponding glides [j, w] when they precede another vowel. This is among the most productive consonantalization rules in the corpus. Additionally, /s/ followed by /i/ + V coalesces into the palatal fricative [ʃ].

Rule 8.1: ia → ja

	Southern Min (Hokkien)	Cebuano
(1)	[ho ⁵³ .piã ⁵³]	[hok ^ɿ pja]
(2)	[dzun ³³ .piã ⁵³]	[dzun ^ɿ pja]
(3)	[li ⁵³ .kiam ²⁴]	[di ^ɿ kjam]
(4)	[kiam ²⁴ .lo ⁵⁵]	[kjam ^ɿ lo]
(5)	[k ^h a ⁵⁵ .lia ^{ɿ4}]	[ka ^ɿ lja ^ɿ]
(6)	[pak ^{ɿ4} .kiau ⁵³]	[pak ^ɿ kjaw]

Rule 8.2: ua → wa

	Southern Min (Hokkien)	Cebuano
(7)	[mi ³³ .suã ²¹]	[mis ^ɿ wa]
(8)	[tau ³³ .kuã ⁵⁵]	[tok ^ɿ kwa]

Rule 8.3: s + i + V → ʃ + V (palatalization)

	Southern Min (Hokkien)	Cebuano
(9)	[sio ⁵⁵ .mai ³³]	[ʃo ^ɿ maj]
(10)	[sio ⁵⁵ .pau ⁵⁵]	[ʃo ^ɿ paw]

Rule 9.1: ui / ue → oj

	Southern Min (Hokkien)	Cebuano
(1)	[ba ^{ɿ32} .tsui ⁵³]	[bat ^ɿ ts ^h oj]
(2)	[ti ⁵⁵ .kue ⁵³]	[ti ^ɿ koj]
(3)	[ɔ ³³ .kue ⁵³]	[u ^ɿ koj]

Rule 9.2: au → aw

	Southern Min (Hokkien)	Cebuano
(4)	[hi ³³ .kau ⁵⁵]	[hi ^ɿ kaw]
(5)	[lau ³³ .lau ³³]	[law ^ɿ law]
(6)	[pak ^{ɿ4} .kiau ⁵³]	[pak ^ɿ kjaw]

Rule 9.3: ai → aj

	Southern Min (Hokkien)	Cebuano
(7)	[peɿ ⁴ .ts ^h ai ²¹]	[p ^h e'ts ^h ay]
(8)	[ka ³³ .i ⁵⁵ .t ^h ai ²⁴]	[ka'tay]

Rule 9.4: au → o or a

	Southern Min (Hokkien)	Cebuano
(9)	[tau ³³ .ge ²⁴]	[to'ge]
(10)	[tau ³³ .iu ²⁴]	['to.jo]
(11)	[tau ³³ .hu ³³]	['ta.ho]

The variation between /o/ and /a/ in Rule 9.4 is not yet fully explained by the present data.

4.3 Vowel Raising (Stressed Syllable)

Based on the corpus, in most cases, mid vowels are raised to high vowels where its syllable is a stressed syllable.

Rule 10.1: ɔ, o → u / (Stressed Syllable)

	Southern Min (Hokkien)	Cebuano
(1)	[ɔ ⁵⁵ .pu ²⁴]	['u.po]
(2)	[ɔ ³³ .kue ⁵³]	['u.koj]
(3)	[hoŋ ³³ .baɿ ³²]	['hum.ba]
(4)	[ko ⁵⁵ .a ⁵³]	['ku.ja]
(5)	[pɔ ⁵³ .t ^h au ²⁴]	['put.t ^h aw]
(6)*	[so ⁵³ .si ²⁴]	[su'si]

Rule 10.2: e → i / (Stressed Syllable)

	Southern Min (Hokkien)	Cebuano
(7)	[ke ⁵⁵ .ti ³³]	['ki.tit]
(8)	[tsu ⁵³ .k ^h eɿ ³²]	[su'ki]
(9)	[he ²⁴ .bi ⁵³]	['hi.be]
(10)	[sin ⁵⁵ .k ^h eɿ ³²]	[siŋ'ki]

Upon the observation of current data, most words follow the prediction. However, there is one counterexample, which indicates that there might be other factors, other than stressed pattern, involved in the mechanism in vowel raising rules, requiring further investigations. Example 10.(6)* shows this clearly: [so'si] would satisfy Shryock's stress prediction just as well as the attested [su'si] does, since both place stress on the final syllable; the only difference is that the unstressed first syllable surfaces with a high vowel [su] rather than a mid vowel [so]. In other words, vowel raising

has applied in an unstressed syllable, which the stress-conditioned formulation of Rule 10 does not predict. This indicates that some factor other than stress assignment is also driving raising in this item, requiring further investigation.

4.4 Vowel Lowering (Unstressed Syllable)

Complementary to Rule 10, high vowels /u/ and /i/ are lowered to /o/ and /e/ in the unstressed syllable. Both the /u/ → /o/ and /i/ → /e/ sub-rules are robust and well-attested.

Rule 11.1: u → o / (Unstressed Syllable)

	Southern Min (Hokkien)	Cebuano
(1)	[bi ⁵³ .hun ⁵³]	['bi.hon]
(2)	[ki ²¹ .su ³²]	['ku.sot]
(3)	[gu ²⁴ .tɔ ³³]	[go'to]
(4)	[tau ³³ .hu ³³]	['ta.ho]
(5)	[hok ²⁴ .bu ³³]	['hok.bo]

Rule 11.2: i → e / (Unstressed Syllable)

	Southern Min (Hokkien)	Cebuano
(6)	[he ²⁴ .bi ⁵³]	['hi.be]
(7)	[a ⁵⁵ .tsi ⁵³]	['a.te]

As with Rule 10, Rule 11 also has counterexamples, drawn from the unstressed portions of items already presented in §4.3. These are listed separately below by target vowel.

Counterexamples to Rule 11.1 (u → o expected, high vowel retained instead)

	Southern Min (Hokkien)	Cebuano
(6)	[so ⁵³ .si ²⁴]	[su'si]
(8)	[tsu ⁵³ .k ^h eɿ ³²]	[su'ki]

Counterexamples to Rule 11.2 (i → e expected, high vowel retained instead)

	Southern Min (Hokkien)	Cebuano
(7)	[ke ⁵⁵ .ti ³³]	['ki.tit]
(10)	[sin ⁵⁵ .k ^h eɿ ³²]	['siŋ.ki]

Example 10.(6) shows vowel raising applying in an unstressed syllable, as already discussed in §4.3. Examples 10.(7), 10.(8), and 10.(10), by contrast, involve an unstressed syllable whose vowel was already high in Southern Min; in all three, the Cebuano output keeps the [+high] specification

rather than lowering it, so the lowering rule simply does not apply. In other words, the stressed syllable requires a high vowel, which may be a licensing effect, but the unstressed syllable does not require a [-high] vowel: it tolerates either. Table 1 summarizes this asymmetry as a 2×2 grid of stress and vowel height.

	[+high]	[-high]
Stressed	allowed (required)	not allowed
Unstressed	allowed	allowed

Table 1: Licensing of vowel height by stress position

4.5 Association of Cebuano Stress Pattern and Vowel Position

According to CEB001, high vowels /i, u/ occur far more frequently than mid vowels /e, o/, suggesting that high vowels are the marked category in Cebuano, while /e/ and /o/ are unmarked. This markedness split also has a historical basis: the native Cebuano vowel system consists only of /i, a, u/, with /e/ and /o/ entering the inventory later, primarily through loanwords (Bunye & Yap 1971; Wolff 1972). The data discussed in Sections 4.3 and 4.4 do not show a simple one-to-one association between vowel markedness and stress, since the unstressed position permits both high and mid vowels (§4.4): a stressed syllable requires a marked high vowel, but an unstressed syllable is not restricted to the unmarked mid vowel. This pattern is visualized in Figure 1, which illustrates the connection between vowel markedness and stress position; the precise mechanism underlying this association, however, remains a question for further research.

Furthermore, the selection of which syllable is stressed is itself unpredictable from the Southern Min source. Shryock’s (1993) research shows that stress preferentially settles on the penultimate syllable when that syllable is heavy, which partly predicts the Cebuano stress pattern; however, this prediction cannot be derived from Southern Min syllable structure alone, because multiple phonological changes apply simultaneously during borrowing, and the Southern Min input attested in present-day dictionaries may differ from the historical input that actually entered Cebuano. For instance, [ke⁵⁵.ti³³] in example 10.(7) has a light penultimate syllable, a case for which Shryock’s theory does not predict stress placement one way or the other; yet the word surfaces in Cebuano as ['ki.tit], with penultimate stress and raising. How this particular outcome arose historically cannot be recovered from the present data, and is a further instance of the unpredictability noted throughout this section.

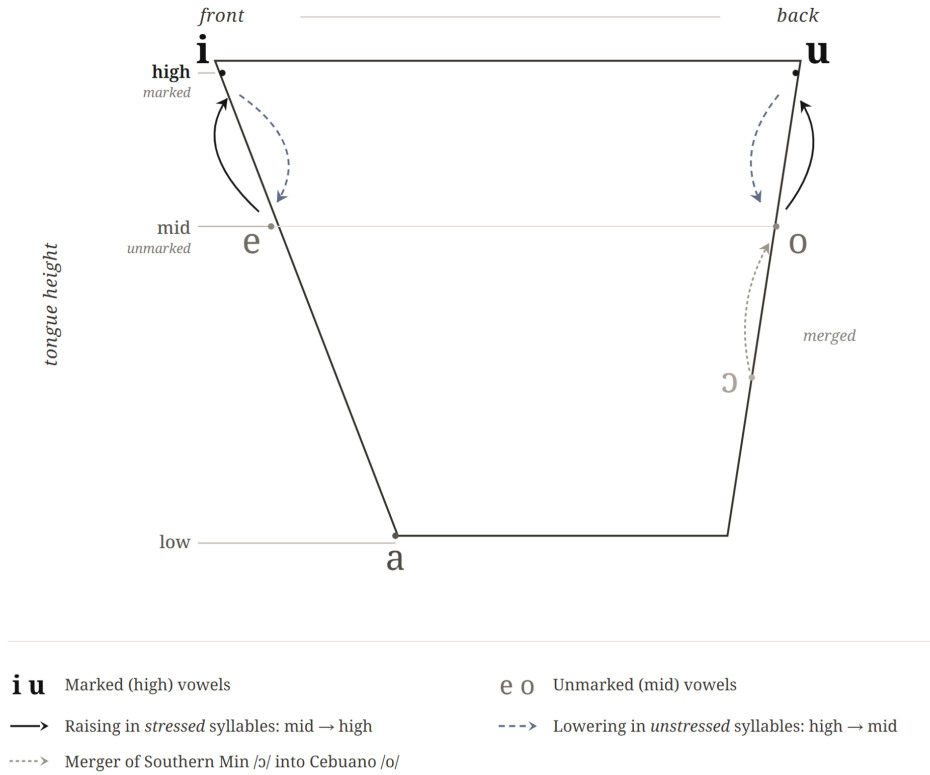


Figure 1: Cebuano vowel space showing markedness and the stress-conditioned height alternation.

5 Results and Discussion: Morpheme-Level Rules

5.1 Morpheme Deletion (Trisyllabic Truncation)

Southern Min trisyllabic words, typically compound morphemes of three Chinese characters, tend to be reduced upon borrowing into Cebuano, likely because Cebuano has a strong preference for disyllabic lexical words. Four examples are attested, displaying different truncation strategies:

	Southern Min (Hokkien)	Cebuano
(1)	[ka ³³ .i ⁵⁵ .t ^h ai ²⁴]	[ka'tay]
(2)	[liam ²⁴ .tɔ ³³ .baʔ ³²]	[ljem'po]
(3)	[p ^h aʔ ³² .si ⁵³ .laŋ ²⁴]	[pas'laŋ]
(4)	[kio ²¹ .kio ²¹ .k ^h au ²¹]	[kjaw'kjaw]

Example 12.(1) shows deletion of the penultimate syllable; example 12.(2) shows deletion of the final syllable. In both cases, the initial syllable is preserved, suggesting a preference for retaining word-initial material. Example 12.(3) involves a more complex process: the vowel of the penultimate

syllable is deleted, and its onset migrates to the coda position of the preceding syllable (syllable fusion). Example 12.(4) involves syllable fusion (/kio/ + /k^hau/ → /kia/ → /kjaw/) followed by reduplication, mirroring the disyllabic reduplicative pattern of the Southern Min source.

The process appears sensitive to the specific syllabic structure and segmental content of the source word, and may also reflect morpheme-level transparency and historical variations. Thus, no consistent deletion rule can be stated.

5.2 Epenthesis (Syllabic Nasal Resolution)

Southern Min has syllabic nasals /sŋ/ and /tŋ/ that form complete syllable without a vowel. Cebuano does not permit syllabic consonants: all syllables must have a vowel. A vowel is therefore inserted to break up the syllabic nasal and produce a well-formed CVC syllable.

Rule 13.1: $\emptyset \rightarrow V / C_ŋ$

	Southern Min (Hokkien)	Cebuano
(1)	[laŋ ²⁴ .sŋ ²⁴]	[lan'soŋ]
(2)	[hue ⁵⁵ .tŋ ²⁴]	[huwe'teŋ]

The selection of the inserted vowel is not yet fully predictable: the closest acoustic approximation of the syllabic nasal /ŋ/ is /eŋ/; however, /sŋ/ → /soŋ/ in 13.(1) and /tŋ/ → /teŋ/ in 13.(2) yield /oŋ/ and /eŋ/ respectively. This variation requires additional data to firm generalization.

5.3 Reduplication

Two items in the corpus surface in Cebuano as reduplicated forms: [kjaw'k-jaw] (already introduced as example 12.(4)) and ['bi.lo-'bi.lo].

The derivation of [kjaw'kjaw] is discussed in §5.1: the Southern Min source [kio²¹.kio²¹.k^hau²¹] first undergoes syllable fusion of its second and third syllables (/kio/ + /k^hau/ → /kia/ → /kjaw/), and only the resulting disyllabic form [kjaw] is reduplicated. The likely motivation for the reduplication step itself is that the Southern Min source word already has reduplicated material in its first two syllables ([kio²¹.kio²¹...]), so the Cebuano output preserves that reduplicative shape even though the specific syllable being copied has changed.

The derivation of ['bi.lo-'bi.lo] 'bilu-bilo' is different. Its Southern Min source is the single (non-reduplicated) form [bi⁵³.lu⁷⁴]. To arrive at ['bi.lo], this source must have undergone checked-tone coda deletion (/t̚/ → ∅, unlike the general pattern of checked-tone coda preservation discussed in §3.6) and vowel lowering (/u/ → /o/, parallel to Rule 11.1), before the resulting disyllabic form was reduplicated to produce ['bi.lo-'bi.lo]. The deletion of the checked-tone coda here, despite its general preservation elsewhere in the corpus, may be conditioned by the reduplication itself, or may instead reflect the dual-route hypothesis (§3.3): the Tagalog form of this word is

bilu-bilo, which has no checked coda to begin with, so if this item entered Cebuano via Tagalog rather than directly from Southern Min, only vowel lowering, and not coda deletion, would need to be posited.

Taken together, both reduplicated items suggest the same general pattern: the source word first undergoes one or more ordinary phonological changes, and only the resulting form is reduplicated afterward. This ordering is not yet established as a general rule, however, and further investigation with additional reduplicated items is needed to confirm it.

6 Rule Summary

Table 2 provides a summary of all fourteen phonological rules documented in Sections 3–5.

Table 2: Summary of Phonological Adaptation Rules (Hokkien → Cebuano)

#	Rule Name	Formalization	Sub-rules
1	Nasalization Loss	$\tilde{V} \rightarrow V$	Rule 1.1
2	Nasal Place Assimilation	$N \rightarrow [\alpha\text{place}] / _ \{ +\text{stop}, \alpha\text{place} \}$	Rules 2.1–2.4
3	Aspiration	$k^h \rightarrow k$; $t^h \rightarrow t^h$; $ts^h \rightarrow ts^h$ (conditional); $p \leftrightarrow p^h$ (unstable)	Rules 3.1–3.4
4	/ts/ Replace- ment	$ts/ts^h \rightarrow s, \int, t, \text{ or } ts^h$ (path-conditioned)	Rules 4.1–4.4
5	Glottal Stop Dele- tion/Insertion	$ʔ \rightarrow \emptyset / _]\sigma$ (coda); $\emptyset \rightarrow ʔ / V_V$ (conditional)	Rules 5.1–5.2
6	Coda Stop Preservation & Insertion	$C^{\text{̣}} \rightarrow C^{\text{̣}}$; $\emptyset \rightarrow C^{\text{̣}} / _ .C$	Rules 6.1–6.2
7	ɔ → o Substi- tution	$\text{ɔ} \rightarrow \text{o}$	Rule 7
8	Glide Forma- tion	$ia/ua \rightarrow ja/wa$; palatal- ization	Rules 8.1–8.3
9	Diphthong Restructur- ing	$ui/ue \rightarrow oj$; $au \rightarrow aw/o$	Rules 9.1–9.4
10	Vowel Rais- ing	$\text{ɔ}/\text{o} \rightarrow \text{u} / \# _$; $e \rightarrow i / _]\sigma$ (stressed)	Rules 10.1– 10.2
11	Vowel Low- ering	$u \rightarrow o / _ C]\sigma$; $i \rightarrow e / _]\sigma$ (unstressed, condi- tional)	Rules 11.1– 11.2
12	Morpheme Deletion	Trisyllabic truncation (no consistent rule)	—

Table 2: Summary of Phonological Adaptation Rules (Hokkien → Cebuano)

#	Rule Name	Formalization	Sub-rules
13	Epenthesis	$\emptyset \rightarrow V / C_n$ (breaks syllabic nasals)	Rule 13.1
14	Reduplication	Disyllabic root reduplicated after prior changes (no consistent rule)	—

Of these fourteen rules, six are unconditional (Rules 1.1, 2.1–2.4, 3.1, 3.2): they apply across the board given the phonological environment. The remaining rules (3.3–3.4, 4.1–4.4, 5.2, 11.2, 12, 14) are conditional, sensitive either to borrowing route or lexical specification, or, in the case of Rule 5.2 and Rules 12/14, simply too sparsely attested in the present corpus to confirm as fully general. The borrowing-route conditionality is a structurally novel hypothesis for Hokkien-Philippine language contact studies and warrants systematic follow-up with larger speaker samples and targeted elicitation.

7 Conclusion

This paper has documented fourteen phonological rules governing the adaptation of Southern Min loanwords into Cebuano, based on fieldwork data from consultant CEB001. The rules span three domains: consonant adaptation, vowel adaptation, and morpheme-level processes.

Several empirical findings are noteworthy. First, Cebuano systematically preserves Hokkien checked-tone coda stops ($/k̚/$, $/t̚/$) where Tagalog categorically deletes them, with the Tagalog coda stop itself undergoing deletion in the same syllable-coda position where the Southern Min source originally had a checked tone. This asymmetric mapping is an instance of what Hsieh et al. (2009) call transphonologization, where a contrast from the source language survives in the borrowing language but is re-encoded onto a different native structure; here, Cebuano keeps the checked-tone codas as unreleased stops, while Tagalog does not. Second, the observed variation in aspiration and affricate replacement is explained by a dual borrowing-route hypothesis: items that entered Cebuano directly from Southern Min preserve aspiration and aspirated affricates, while items that entered via Tagalog lost aspiration and underwent $/ts/$ simplification. Third, vowel raising and lowering are conditioned by stress position, a finding that connects the loanword phonology to the native Cebuano stress system described in Shryock (1993), and which further suggests an asymmetric licensing relationship between stress and vowel markedness (§4.5).

Several questions remain open. The counterexample to the vowel raising rule, and the related cases in which an unstressed syllable retains a high vowel instead of lowering (§4.4), are not yet fully explained; stress position is a strong predictor overall, but the present dataset cannot rule

out an additional conditioning factor. The epenthetic vowel selection in syllabic nasal resolution (Rule 13.1) also lacks a principled explanation. The absence of /p̚/ coda stops in the corpus may reflect a real phonotactic gap or may be a result of the limited sample. Future work should include additional elicitation sessions with CEB001, acceptability judgment tasks using artificial words, and data from additional Cebuano and Tagalog speakers and dialects to assess the generalizability of the present findings.

More broadly, this study contributes to a growing body of work on Southeast Asian Hokkien loanword phonology, joining comparable analyses of Indonesian (Aryanti 2021) and Tagalog (Chan-Yap 1977). The comparison of Cebuano with both Indonesian and Tagalog reveals that the phonological adaptation of Hokkien loanwords is language-specific even within geographically and historically related contact communities. What is preserved in one language may be deleted in another, and the specific adaptation strategies reflect the unique phonological grammar of each receiving language, as mathematical functions with conditional constraints, rather than a universal adaptation template.

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AI Usage Report

Claude (Anthropic) was used as a research and drafting assistant throughout the preparation of this paper, under the author’s direction and review.

Use fell into five areas:

- **Grammar & Polish** — Claude reviewed throughout the paper for grammar, clarity, and academic register, and proposed phrasing edits that the author then accepted, rejected, or revised.
- **IPA Cross-Checking Against Online and Print Dictionaries** — Claude assisted in matching and verifying Southern Min IPA transcriptions against the Ministry of Education Taiwanese Hokkien online dictionary (sutian.moe.edu.tw), the Revised Formosan Pronunciation Dictionary (chhoe.taigi.info), itaigi.tw, and taigi.fhl.net, and in cross-checking Cebuano IPA forms against Binisaya.com, Wiktionary, and Wolff (1972)'s *A Dictionary of Cebuano Visayan*. All matches were reviewed by the author against CEB001's elicited productions; the dictionaries served as a verification aid rather than a primary data source.
- **Formatting** — Claude assisted with table formatting, rule numbering consistency across sections, reconciling the in-text corpus data with the Appendix A wordlist, and conversion of the paper into the present Manchester Phonology Meeting (MfM) $\text{\LaTeX}$ template.

Appendix A: Hokkien-Cebuano-Tagalog Corpus

Table 3 presents the full corpus of 65 lexical items underlying the analysis in Sections 3–5. The Cebuano IPA column has been reconciled with the transcriptions used in the discussion sections above; where the two differed, the form used in the Results and Discussion sections was treated as authoritative.

Table 3: Hokkien-Cebuano-Tagalog Corpus (CEB001 data)

S. Min (Hokkien)	Cebuano	Tagalog	Char.	Gloss
[bin ³³ .pɔ ²¹]	[bim'po]	bimpo	面布	face/wash towel
[bak ¹⁴ .k ^h ia ¹⁴]	[bak'kia]	bakiya	木屐	wooden clogs
[hĩ ³³ .kau ⁵⁵]	[hi'kaw]	hikaw	耳鉤	earrings
[lau ³³ .lau ³³]	[law'law]	lawlaw	老老	loose, hanging
[so ⁵³ .si ²⁴]	[su'si]	susi	鎖匙	key
[siŋ ⁵⁵ .li ⁵³ .laŋ ²⁴]	[saŋ'laŋ]	sanglay	生理人	Chinese trader (archaic)
[hia ¹⁴]	[hi'ja]	hiya	額	shame, embarrassment
[sin ⁵⁵ .k ^h e ¹³²]	[siŋ'ki]	singki	新客	newcomer; Chinese-looking person
[kio ²¹ .kio ²¹ .k ^h au ²¹]	[kjaw'kjaw]	kiyaw-kiyaw	叫叫哭	fretting, fussing
[p ^h a ¹³² .si ⁵³ .laŋ ²⁴]	[pas'laŋ]	paslang	拍死人	to kill, murder
[bo ²⁴ .ui ⁵⁵ .sit ¹⁴]	[buwi'sit]	buwisit	無衣食	annoying person; bad luck
[sue ⁵⁵ .a ⁵⁵]	[su'jaʔ]	suya	衰啊	jealousy, envy
[tik ³² .k ^h ak ³²]	[ti'jak]	tiyak	的確	sure, certain
[ke ⁵⁵ .ti ³³]	['ki.tit]	kiti	雞弟	chick, young fowl
[lau ³³ .iŋ ⁵⁵]	['law'in]	lawin	老鷹	hawk, falcon
[t ^h aŋ ²⁴ .a ⁵³]	[t ^h aŋ'a]	tanga	蟲仔	stupid; bug/worm
[li ⁵³ .kiam ²⁴]	[di'kjam]	dikiyam	李鹹	salted dried plums
[gu ²⁴ .tɔ ³³]	[go'to]	goto	牛肚	ox/cow tripe (dish)
[liam ²⁴ .tɔ ³³ .ba ¹³²]	[ljem'po]	liyempo	liam肚肉	pork belly
[pe ¹⁴ .ts ^h ai ²¹]	[p ^h e'ts ^h ay]	petsay	白菜	Chinese cabbage, pechay
[ts ^h i ⁵⁵ .tau ³³]	[si'taw]	sitaw	青豆	string beans
[ti ⁵⁵ .tɔ ³³]	['ti.to]	tito	豬肚	pork stomach
[tau ³³ .ge ²⁴]	[to'ge]	toge	豆芽	bean sprouts
[ɔ ⁵⁵ .pu ²⁴]	['u.po]	upo	烏匏	bottle gourd
[ba ¹³² .tsui ⁵³]	[bat'ts ^h oj]	batsoy	肉水	pork offal soup
[bi ⁵³ .hun ⁵³]	['bi.hon]	bihon	米粉	rice vermicelli
[bi ⁵³ .ko ⁵⁵]	[bi'ko]	biko	米糕	sticky rice cake
[bi ⁵³ .lut ¹⁴]	['bi.lo-'bi.lo]	bilu-bilo	米糰	rice balls in ginataan
[hok ⁵³ .piã ⁵³]	[hok'pja]	hopiya	好餅	mung-bean cake
[hoŋ ³³ .ba ¹³²]	['hum.ba]	humba	燜肉	braised pork dish
[he ²⁴ .bi ⁵³]	['hi.be]	hibe	蝦米	dried shrimp

Table 3: Hokkien-Cebuano-Tagalog Corpus (CEB001 data)

S. Min (Hokkien)	Cebuano	Tagalog	Char.	Gloss
[kiam ²⁴ .lo ⁵⁵] [lo ⁵⁵ .mi ³³]	[kjam'lo] ['lo.mi]	kiyamlo lome	鹹滷 滷麵	stewed/braised dish thick noodle dish, lomi
[dzun ³³ .piã ⁵³] [baʔ ³² .mi ³³] [mi ³³ .ki ⁵⁵] [mi ³³ .suã ²¹] [pian ³³ .sit ⁴] [sio ⁵⁵ .mai ³³]	[dzun'pja] ['ma.mi] [mi'ki] [mis'wa] [pan'sit] [ʃo'maj]	lumpiya mami miki misuwa pansit siyomay	潤餅 肉麵 麵枝 麵線 便食 燒賣	spring roll noodle soup egg noodles fine flour noodles noodle dish steamed dumpling, siomai
[sio ⁵⁵ .pau ⁵⁵] [tau ³³ .hu ³³] [tau ³³ .sĩ ³³] [ti ⁵⁵ .kue ⁵³] [tau ³³ .kuã ⁵⁵] [tau ³³ .iu ²⁴] [ɔ ³³ .kue ⁵³] [bi ⁵³ .lau ³³] [laŋ ²⁴ .sɿ ²⁴] [k ^h a ⁵⁵ .liaʔ ⁴]	[ʃo'paw] ['ta.ho] [taw'si] [ti'koj] [tok'kwa] ['to.jo] [u'koj] [bi'laʔ.o] [lan'soŋ] [ka'ljaʔ]	siyopaw taho tawsi tikoy tokuwa toyo ukoy bilao lansong kaliya	燒包 豆腐 豆豉 甜粿 豆乾 豆油 芋粿 米簍 籠床 跤箝	steamed bun, siopao tofu pudding dessert salted black beans glutinous rice cake firm tofu soy sauce shrimp fritter winnowing tray bamboo steamer large bamboo tray/basket
[a ⁵⁵ .tsi ⁵³] [dzi ³³ .tsi ⁵³]	['a.te] [di'ts ^h e]	ate ditse	阿姊 二姊	elder sister (addr.) second elder sister (addr.)
[ko ⁵⁵ .a ⁵³] [dzi ³³ .ko ⁵⁵]	['ku.ja] [di'ko]	kuya diko	哥仔 二哥	elder brother (addr.) second elder brother (addr.)
[hok ⁴ .bu ³³] [ki ²¹ .sut ³²] [pɔ ⁵³ .t ^h au ²⁴] [gin ²⁴ .tsam ⁵³] [taŋ ²⁴ .soʔ ³²] [ka ³³ .i ⁵⁵ .t ^h ai ²⁴]	['hok.bo] ['ku.sot] ['put.t ^h aw] [gin'sam] [tan'so] [ka'tay]	hukbo kusot puthaw gintsam tanso katay	服務 鋸屑 斧頭 銀斬 銅索 共創	army, troops sawdust iron; small axe smith's chisel copper, bronze to butcher, cut up meat
[pak ⁴ .kiau ⁵³] [tsu ⁵³ .k ^h eʔ ³²]	[pak'kjaw] [su'ki]	pakyaw suki	縛繳 主客	wholesale; contract regular customer/vendor
[hue ⁵⁵ .tɿ ²⁴]	[huwe'teŋ]	huweteng	花棟	illegal numbers game, jueteng
[koŋ ²¹] [poŋ ³³] [toŋ ⁵⁵] [tsui ⁵³ .kui ⁵³]	['kaŋ] ['poŋ] ['ton] [ʃo'koy]	kang pong ton siyukoy	槓 碰 東 水鬼	mahjong term (kang) mahjong term (pong) gambling house cut merman/sea spirit